

VIX Tupiniquim II: Non-Linear Modeling and Interpretability of Financial Stress via Gradient Boosting Machines and SHAP Values

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Abstract

This paper extends the methodological framework of the “VIX Tupiniquim” by proposing a non-linear model of domestic financial stress using Gradient Boosting Machines (GBM). To ensure economic discipline in the presence of limited macroeconomic sample sizes, the model architecture incorporates monotonic sign constraints based on economic theory and structural interaction constraints across macroeconomic blocks. The hierarchical importance and directional contribution of macro-financial determinants to the model’s predictions are evaluated using Cooperative Game Theory (*SHAP Values*). Finally, the dynamics of the proposed indicator are compared with the Economic Policy Uncertainty Index (EPU) through temporal precedence tests (Granger Causality). The empirical results show that the non-linear approach tracks the major financial stress episodes dated by CODACE/FGV without exhibiting statistically detectable linear temporal precedence with respect to news-based economic uncertainty.

Keywords: Financial Stress, Non-Linearity, Gradient Boosting Machine, XGBoost, SHAP Values, Monotonic Constraints, VIX Tupiniquim.

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1 Introduction

Tracking financial stress and systemic risk in emerging economies involves non-linear dynamics, asymmetric responses across the business cycle, and complex transmission mechanisms across markets. In the first paper of this series (*VIX Tupiniquim I: Building a Synthetic Financial Stress Indicator via PCA and Linear Regularization*)¹, the linear dimension was explored using Principal Component Analysis (PCA) and Regularization Methods (LASSO, Ridge, and Elastic Net). While effective for dimensional reduction and parsimonious feature selection, linear formulations impose rigid assumptions of additivity and constant elasticities over time.

In this exercise, we develop the **VIX Tupiniquim II**, adopting a predictive core based on Gradient Boosting Machines (GBM), computationally implemented via Extreme Gradient Boosting (XGBoost). The primary goal is to capture non-linearities and behavioral shifts in the relationship between macro-financial covariates and macroeconomic conditions, mitigating “black-box” opacity while preserving conceptual consistency.

To enforce economic discipline, monotonic sign constraints on regressors and structural interaction constraints between transmission channels are embedded directly into the tree-building process. Model interpretability is conducted using SHAP values (*SHapley Additive exPlanations*), evaluating the marginal contribution of each variable to the estimated index. The indicator can be interpreted as a synthetic measure of financial stress conditioned on the non-linear relationship between observed macro-financial conditions and business confidence. The term *VIX Tupiniquim* serves as a conceptual analogy to a domestic financial stress gauge, without methodological equivalence to the CBOE VIX.

2 Methodology: Non-Linear Model Architecture

The construction of the indicator relies on supervised machine learning with theoretical constraints. The framework is structured around the definition of the target variable, the constrained optimization algorithm, and the prediction attribution method.

2.1 Target Definition (y) and Objective Function

The supervised algorithm uses the seasonally adjusted series of the **Business Confidence Index (ICE/FGV)** as the target variable (y_t), obtained after applying an STL structural decomposition filter (ICE_{SA}). The ICE serves as a broad proxy for aggregate economic sentiment in Brazil.

The XGBoost model predicts business confidence $\hat{y}_t = f(\mathbf{x}_t)$ conditional on a vector of lagged explanatory features $\mathbf{x}_t$. Training minimizes the regularized loss objective:

$$\mathcal{L}(\phi) = \sum_{t=1}^n (y_t - \hat{y}_t)^2 + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

where $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2$ penalizes tree complexity (number of terminal leaves T and leaf weights ω).

Because an increase in financial stress corresponds to a deterioration in business confidence, the model’s predicted output ($\hat{y}_t$) is inverted and standardized to generate an index centered at base

¹For complete methodological details of the previous work, click [here](#).

100:

$$\text{VIX}_{\text{raw},t} = -1 \times \left(\frac{\hat{y}_t - \bar{\hat{y}}}{\sigma_{\hat{y}}} \right) \quad (2)$$

$$\text{VIX Tupiniquim } \Pi_t = 100 + 10 \times \text{VIX}_{\text{raw},t} \quad (3)$$

where $\bar{\hat{y}}$ and $\sigma_{\hat{y}}$ denote the sample mean and standard deviation of model predictions.

2.2 Monotonic and Interaction Structural Constraints

Given the modest sample size typical of monthly macroeconomic time series ($N = 162$), unconstrained tree-based algorithms can fit spurious sample-specific correlations. To regularize the learning process, two structural constraints were incorporated into the model architecture:

1. **Monotonic Constraints** ($m_j \in \{-1, 0, 1\}$): Tree splits are constrained to ensure that the estimated partial derivative between covariate x_j and confidence $\hat{y}$ has an invariant sign:

$$\frac{\partial \hat{y}}{\partial x_j} \geq 0 \quad \text{if } m_j = +1 \quad (\text{Ibovespa Return, B3 Volume, and Yield Curve Slope}) \quad (4)$$

$$\frac{\partial \hat{y}}{\partial x_j} \leq 0 \quad \text{if } m_j = -1 \quad (\text{Selic, CDS, NPL, Spreads, FX Volatility, and Global VIX}) \quad (5)$$

2. **Structural Interaction Constraints:** Covariates are partitioned into two conceptual blocks:

- *Block 1 (Sovereign Risk, Monetary Policy, and External Shocks):* Selic Target Rate, Brazil 5Y Sovereign CDS, FX Realized Volatility, and Global CBOE VIX Index;
- *Block 2 (Domestic Credit and Capital Markets):* Banking Spreads (Households and Corporations), Non-Performing Loans (NPL), Yield Curve Slope, B3 Trading Volume, and Ibovespa Return.

The algorithm is restricted to split nodes only among variables within the same block. This structural constraint serves as a simplifying assumption to preserve the functional independence of transmission channels.

2.3 Interpretability via Game Theory (SHAP Values)

To measure the marginal contribution of each regressor to the predicted output, we employ SHAP values (*SHapley Additive exPlanations*), rooted in Cooperative Game Theory (*Shapley, 1953*). The SHAP value ($\phi_i(x)$) decomposes the prediction into additive feature attributions:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (6)$$

where F is the complete set of features, S is a subset excluding variable i , and $f_x(S)$ is the model prediction evaluated on coalition S . SHAP values provide an additive decomposition of marginal contributions in predictive space and do not imply structural causal identification.

3 Data

The empirical dataset and optimal lag selection using the Bayesian Information Criterion (BIC) relative to the ICE follow the protocol established in *VIX Tupiniquim I*, covering the period from November 2011 to April 2026 (162 synchronized monthly observations). Complete descriptions of data sources (Central Bank of Brazil, FGV, CBOE, B3, and Investing.com) and STL decomposition procedures are available in the previous study.

4 Construction and Interpretation of VIX Tupiniquim II

Hyperparameter tuning was conducted via *Grid Search* with sequential temporal validation (*TimeSeriesSplit*), preserving chronological ordering. The optimal configuration selected: maximum depth (*max_depth*) of 3, learning rate (*learning_rate*) of 0.05, and subsampling ratio (*subsample*) of 0.8.

4.1 Ablation Study: Predictive Cost of Structural Constraints

To quantify the predictive cost of imposing theoretical constraints, an ablation study was conducted comparing the constrained model against an unconstrained benchmark estimated under identical hyperparameters. Performance was evaluated using a 5-fold expanding window cross-validation (*TimeSeriesSplit*), reported in Table 1.

Table 1: **Ablation Study: Predictive Performance With and Without Structural Constraints**

Model Specification	RMSE (CV)	MAE (CV)	R^2 (CV)
Unconstrained XGBoost (<i>Benchmark</i>)	9.2057	6.8494	−6.5615
Constrained XGBoost (Structural Architecture)	9.7031	7.5538	−7.6462

Source: Author’s elaboration. Metrics represent out-of-sample averages across sequential cross-validation folds ($n_{\text{splits}} = 5$). The negative out-of-sample R^2 values stem from the strong persistence of the ICE series and heavy penalization during early, shorter folds of the *TimeSeriesSplit* (where training fold means differ from subsequent regime shifts), serving as a strictly relative comparative metric across specifications.

Imposing structural constraints results in a modest loss of predictive performance, with a 5.4% increase in RMSE and a 10.3% increase in MAE relative to the unconstrained benchmark, consistent with the greater functional flexibility of the unconstrained model. In return, the constrained specification incorporates *a priori* economic discipline, enforcing the specified monotonic directions and block partitions out-of-sample.

4.2 Relative Feature Importance vs. SHAP Predictive Contributions

The internal architecture of the model is evaluated using two complementary metrics:

1. **Relative Feature Importance (Figure 1):** Measures the relative gain (*Feature Importance Gain*) of each regressor in reducing squared error across tree splits, without directional attribution.

2. **Mean Prediction Contribution (Figure 2):** Quantifies the mean absolute magnitude of contributions to prediction ($\text{mean}(|\text{SHAP value}|)$). Color indicates the average directional contribution: red represents an association with reduced estimated confidence (increased stress), while green indicates an association with higher estimated confidence (attenuated stress).

Figure 1: **Relative Feature Importance (Gain)**

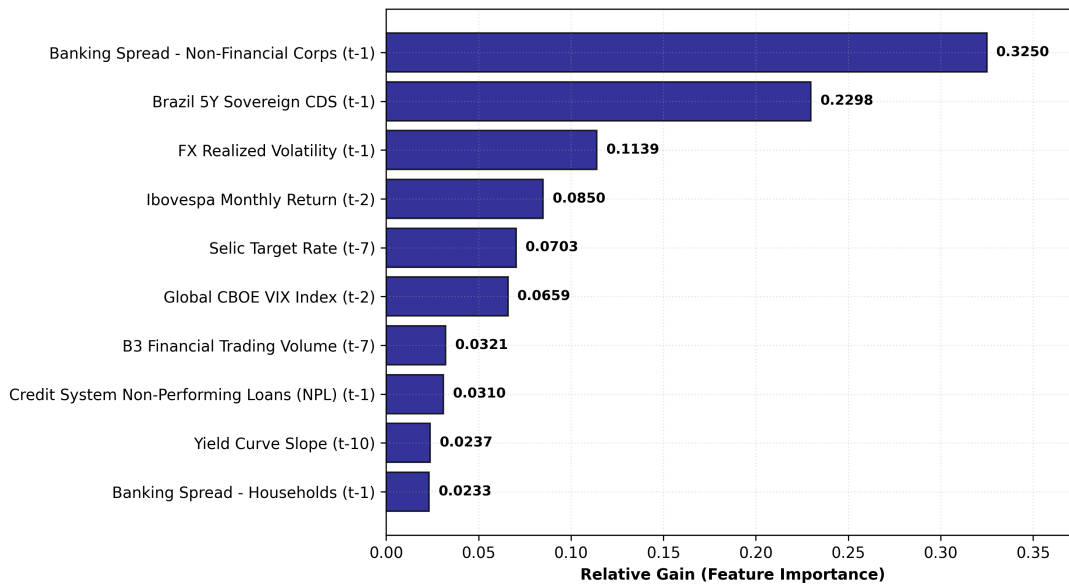
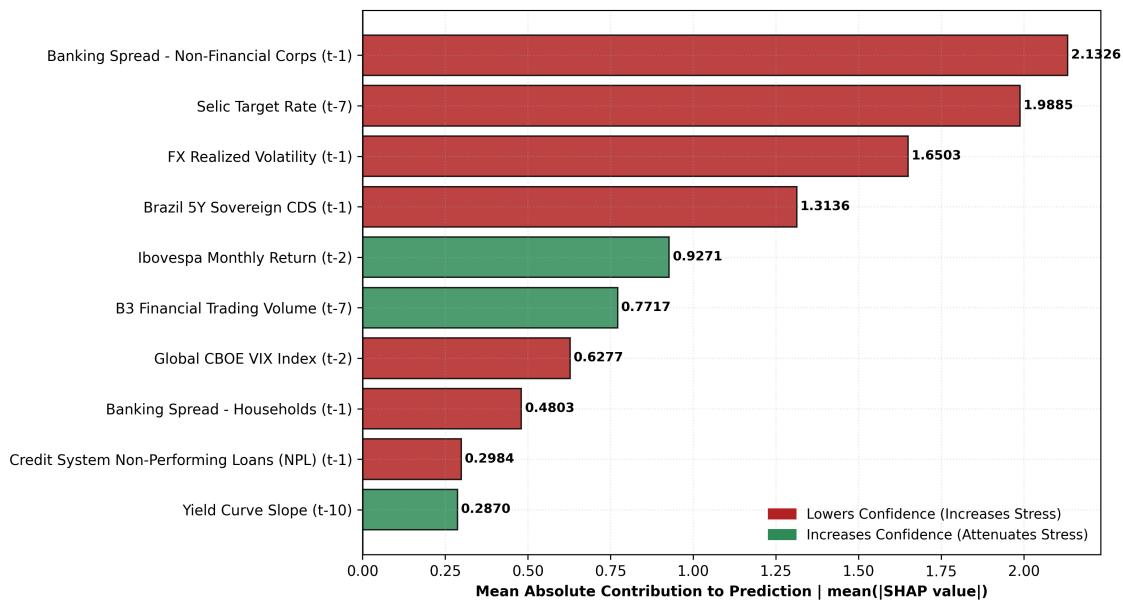


Figure 2: **Mean Absolute Contribution to Prediction (SHAP)**



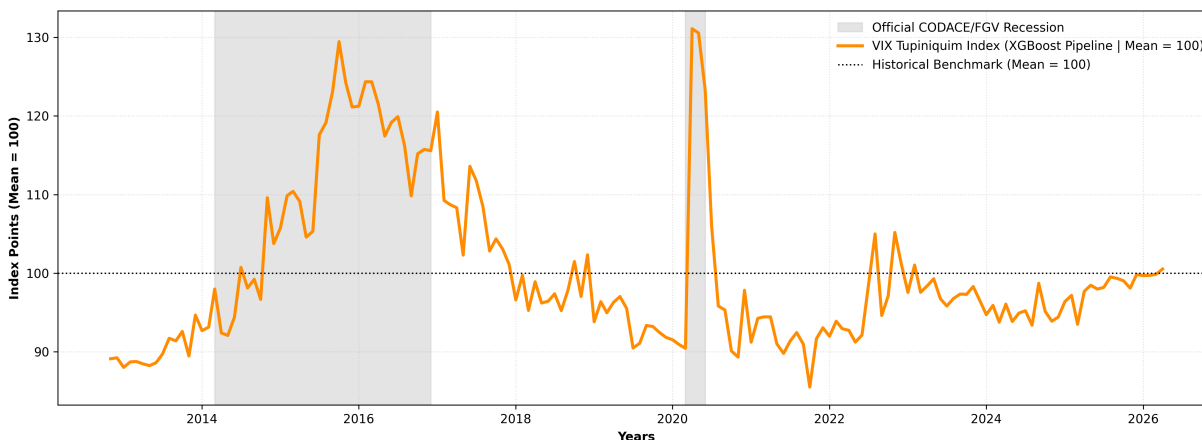
The joint analysis of Figures 1 and 2 reveals the following predictive patterns:

- **Credit Frictions and Funding Costs:** Banking Spread - Non-Financial Corps ($t-1$) and Selic Target Rate ($t-7$) stand out as the primary drivers of lower estimated confidence, exhibiting high partition gain (0.3250 and 0.0703) and the highest mean absolute SHAP contributions (2.1326 and 1.9885), strongly contributing to financial stress.
- **Sovereign Risk and FX Transmission:** Brazil 5Y Sovereign CDS ($t-1$) and FX Realized Volatility ($t-1$) account for notable tree gains (0.2298 and 0.1139) and strong SHAP contributions (1.3136 and 1.6503), consistently associating with confidence contractions (red bars).
- **Liquidity and Capital Market Indicators:** Ibovespa Monthly Return ($t-2$) (0.9271), B3 Financial Trading Volume ($t-7$) (0.7717), and Yield Curve Slope ($t-10$) (0.2870) show positive predictive contributions (green bars), capturing periods of liquidity and lower financial tension.

4.3 The Historical Series of VIX Tupiniquim II

The historical trajectory of the **VIX Tupiniquim II** is shown in Figure 3.

Figure 3: **Historical Series of VIX Tupiniquim II — Non-Linear XGBoost Model**



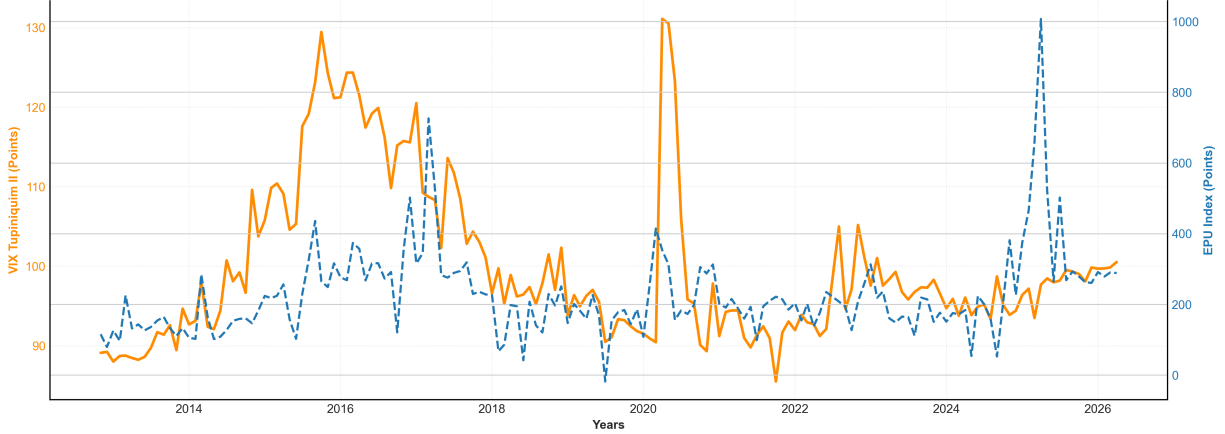
The indicator tracks key historical stress and contraction episodes dated by CODACE/FGV:

- **2014–2016 Recession:** The indicator showed a persistent rise during the recessionary period, exceeding 120 points during its acute phase, reflecting fiscal deterioration, tightening credit conditions, and political crisis developments.
- **2020 Pandemic Shock:** The model captured the acute nature of the shock, recording a spike near 138 points at the onset of the health crisis, followed by a rapid reversal of the indicator.
- **2023–2026 Period:** The index oscillates around its historical baseline (100 points), reflecting the balance between restrictive interest rates and relatively stable sovereign risk premia.

5 Temporal Precedence Analysis with Brazil EPU

To examine whether the synthetic financial stress index exhibits redundancy or temporal precedence relative to news-based economic policy uncertainty, we compare VIX Tupiniquim II against the *Economic Policy Uncertainty Index* (Brazil EPU) developed by *Baker, Bloom and Davis* (Figure 4).

Figure 4: **VIX Tupiniquim II and Brazil EPU — Seasonally Adjusted Series**



Both series are stationary in levels ($I(0)$) according to the Augmented Dickey-Fuller (ADF) test, with p -values of 0.0122 for VIX Tupiniquim II and 0.0036 for the seasonally adjusted EPU (STL). Table 2 reports the Granger causality test results:

Table 2: **Granger Causality Test in Levels: VIX Tupiniquim II vs. Brazil EPU**

Precedence Relationship (H_0)	Lag (k)	F -Statistic	p -value	Decision ($\alpha = 5\%$)
VIX Tupiniquim $\nrightarrow$ EPU (VIX does not Granger-cause EPU)	1	2.2741	0.1337	Fail to Reject H_0
	2	1.0147	0.3651	Fail to Reject H_0
	3	0.3799	0.7674	Fail to Reject H_0
EPU $\nrightarrow$ VIX Tupiniquim (EPU does not Granger-cause VIX)	1	2.9712	0.0869	Fail to Reject H_0
	2	1.5827	0.2088	Fail to Reject H_0
	3	1.1468	0.3328	Fail to Reject H_0

Note: As an additional robustness check for autoregressive time series, the Toda and Yamamoto (1995) procedure was estimated via $\text{VAR}(k + d_{\max})$ with $k = 1$ and $d_{\max} = 1$. The modified Wald test confirmed the absence of statistically significant precedence in both directions: $\text{VIX} \rightarrow \text{EPU}$ (Wald = 0.3578; p -value = 0.5498) and $\text{EPU} \rightarrow \text{VIX}$ (Wald = 1.0833; p -value = 0.2980).

The test fails to reject the null hypothesis of non-causality across all evaluated lag specifications.

5.1 Interpretation of Informational Dynamics

One plausible interpretation for the lack of mutual linear precedence lies in the distinct nature of the underlying information channels:

1. **Media Narrative Uncertainty (EPU):** Derived from keyword frequency in major newspaper articles (*Folha de S. Paulo*), capturing the intensity of political debate and regulatory proposals. Media coverage spikes do not necessarily imply immediate credit contractions or portfolio reallocation.
2. **Market Pricing and Balance Sheet Constraints (VIX Tupiniquim):** Reflects interest rates, exchange rate volatility, banking spreads, and sovereign CDS spreads, capturing the effective cost of capital and observed financial constraints.

Consequently, the proposed indicator operates as a complementary metric to narrative uncertainty measures, focusing specifically on aggregate financial conditions.

6 Concluding Remarks

The development of the VIX Tupiniquim II using Gradient Boosting Machines with structural constraints demonstrates how non-linear machine learning techniques can be disciplined by economic theory in small macroeconomic samples. Embedding monotonic constraints and analyzing predictions via SHAP values enables mapping financial stress while maintaining economic coherence. The comparison with the EPU index found no evidence of linear temporal precedence in either direction, suggesting that the synthetic indicator offers a complementary dimension to news-based uncertainty metrics, serving as a structured barometer for monitoring domestic macro-financial risk.